

CSA1509 – Cloud Computing and Big Data Analytics

EXPERIMENT - 1: Flight Reservation System

Theory: Create a simple cloud software application for Flight Reservation System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as one way, round trip, From, to, Departure, Departure, Travelers & Class, Fare Type, International and Domestic Flights etc.

Aim: To create a simple cloud software application for a Flight Reservation System using a Cloud Service Provider (Zoho Creator) to demonstrate the Software as a Service (SaaS) model.

Algorithm:

1. Login to the Cloud Service Provider (Zoho Creator) and create a new application named "Flight Reservation System".
2. Design a form with the necessary fields such as Passenger Name, Mobile Number, Trip Type (One Way/Round Trip), Flight Type (Domestic/International), From, To, Departure Date, Return Date, Travelers, Class, Fare Type and Payment Mode.
3. Publish the application and enter sample passenger details to test the booking process.
4. View the submitted records in the auto-generated report section to verify that the booking data has been stored correctly.

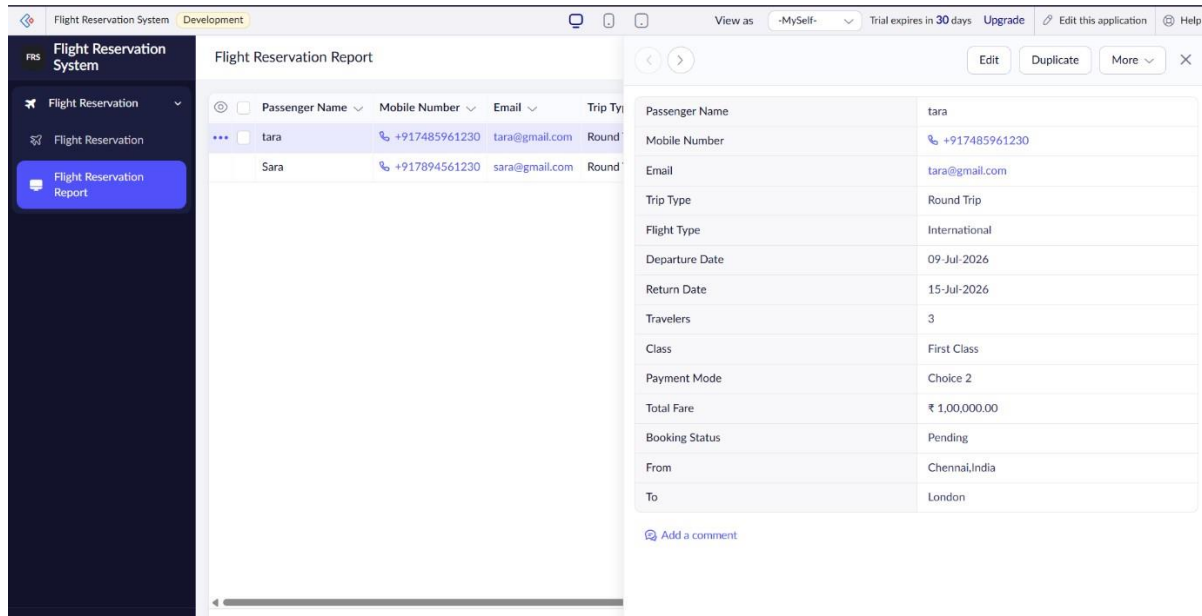
Result: A cloud-based Flight Reservation System was successfully created and deployed using Zoho Creator, and the SaaS model was demonstrated by entering and retrieving sample flight booking data.

Screenshots:

The screenshot displays the 'Flight Reservation System' form within the Zoho Creator interface. The form is titled 'Flight Reservation' and includes the following fields:

- Passenger Name *
- Mobile Number * (with a dropdown for country code, currently showing +91 and a text input for the number 81234 56789)
- Email *
- Trip Type * (with radio buttons for One Way and Round Trip)
- Flight Type * (with radio buttons for Domestic and International)
- From *
- To *
- Departure Date * (with a date picker showing dd-MMM-yyyy)
- Return Date * (with a date picker showing dd-MMM-yyyy)
- Travelers * (with a text input showing #####)
- Class * (with a dropdown menu showing -Select-)
- Payment Mode (with a dropdown menu showing -Select-)

The Zoho Creator interface shows the application is in 'Development' mode, with a trial period of 18 days. The sidebar on the left contains links for 'Flight Reservation' and 'Flight Reservation Report'.



EXPERIMENT - 2: Property Buying & Rental System

Theory: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement, Rental Agreement, Area, place, Range, property value etc.

Aim: To create a simple cloud software application for a Property Buying & Rental process (in Chennai city) using a Cloud Service Provider to demonstrate the SaaS model.

Algorithm:

1. Create a new application on the Cloud Service Provider named "Property Buying & Rental System".
2. Design a form with fields such as Customer Name, Buy/Rent, Property Type, Area (Square Feet), Place, Budget Range, Property Value, Rental Agreement, Rental Period, Owner Name, Address and Status.
3. Publish the form and submit sample property records to test the buying and rental workflow.
4. Verify the stored records in the report section, confirming successful data capture.

Result: A cloud-based Property Buying & Rental application for Chennai city was successfully built and deployed, and sample property records were stored and displayed correctly, demonstrating SaaS.

Screenshots:

abc - Property Buying & Rental System

Ask Gemini

creatorapp.zoho.in/abcportal66/environment/development/property-buying-rental-system/#Form:Property_Booking

Property Buying & Rental SystemDevelopmentView as -MySelf- Trial expires in 18 days Upgrade Edit this application Help

Property Buying & Rental System

Property Booking

Property Booking

Property Booking Report

Property Booking

Customer Name *

First NameLast Name

Mobile Number *

+91 81234 56789

Email *

Buy / Rent *

BuyRent

Property Type *

-Select-

Area (Square Feet) *

#####

Place *

-Select-

Budget Range *

-Select-

Property Value

###,###,##

Rental Agreement *

-Select-

Rental Period *

#####

Owner Name *

Property Buying & Rental SystemDevelopmentView as -MySelf- Trial expires in 18 days Upgrade Edit this application Help

Property Buying & Rental System

Property Booking

Property Booking

Property Booking Report

Property Booking

Property Type *

-Select-

Area (Square Feet) *

#####

Place *

-Select-

Budget Range *

-Select-

Property Value

###,###,##

Rental Agreement *

-Select-

Rental Period *

#####

Owner Name *

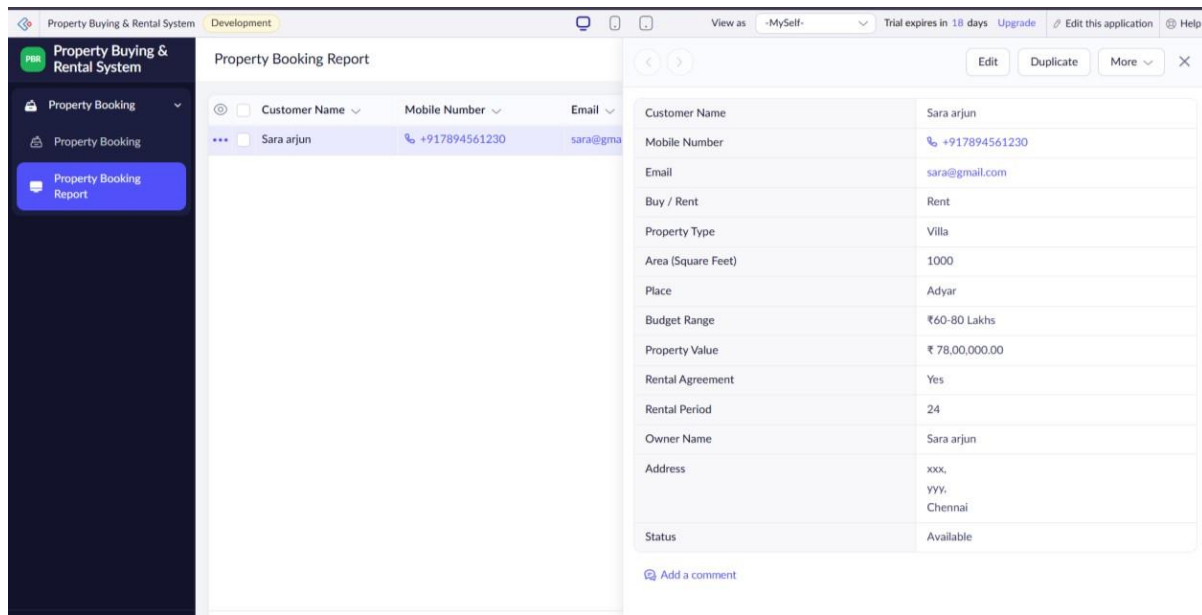
First Name

Address *

Status *

-Select-

Cancel



EXPERIMENT - 3: Car Booking Reservation System

Theory: Create a simple cloud software application for Car Booking Reservation System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as search for cabs, from, to, rental, out station, Package type, hours/Days etc.

Aim: To create a simple cloud software application for a Car Booking Reservation System using a Cloud Service Provider to demonstrate SaaS.

Algorithm:

1. Create a new application named "Car Booking" on the Cloud Service Provider.
2. Design a form with fields such as Name, Employee ID, Mobile Number, Gender, Booking Date, Travel Date, Pickup Time, Pickup Address and Destination Address, including rental, out-station and package type/hours-days options.
3. Publish the application and enter sample booking details using the "Book Now" action.
4. Check the Bookings report section to confirm the cab booking details were recorded successfully.

Result: A cloud-based Car Booking Reservation System was successfully developed and deployed, and a sample cab booking was created and verified in the report, demonstrating the SaaS model.

Screenshots:

Car Booking

Book a Cab

Bookings

Book a Cab

Name *

First NameLast Name

Employee ID *

Email Id *

Mobile Number *

+91

81234 56789

Gender *

Male

Female

Booking Date *

21-Jul-2026

Travel Date *

dd-MMM-yyyy

Pickup Time *

-Select-

Pickup Address *

Address Line 1

Address Line 2

Car Booking

Book a Cab

Bookings

Book a Cab

Pickup Time *

-Select-

Pickup Address *

Address Line 1

Address Line 2

City / DistrictState Province

-Select-

Postal CodeCountry

Destination Address *

Address Line 1

Address Line 2

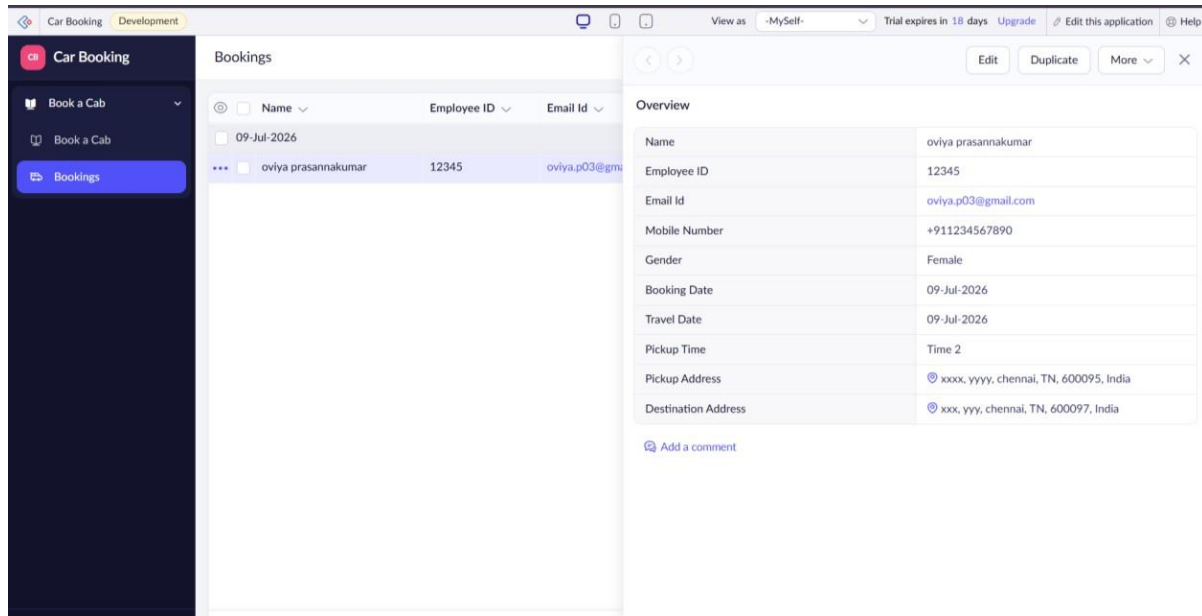
City / DistrictState Province

-Select-

Postal CodeCountry

Book Now

Reset



EXPERIMENT - 4: Library Book Reservation System (SIMATS Library)

Theory: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Aim: To create a simple cloud software application for a Library Book Reservation System for SIMATS Library using a Cloud Service Provider to demonstrate SaaS.

Algorithm:

1. Create a new application named "SIMATS Library Book Reservation System" on the Cloud Service Provider.
2. Design a form with fields such as Student Name, Register Number, Department, Title of the Book, Author, Publication, Year, Edition, Number of Copies, Rack Number, Book Status, Issue Date, Return Date and Librarian Name.
3. Publish the form and submit a sample book reservation record to test the workflow.
4. Verify the submitted record in the Library Book Reservation Report to confirm the status (taken/returned) is tracked correctly.

Result: A cloud-based Library Book Reservation System for SIMATS Library was successfully created, and a sample book issue/return record was stored and verified, demonstrating SaaS.

Screenshots:

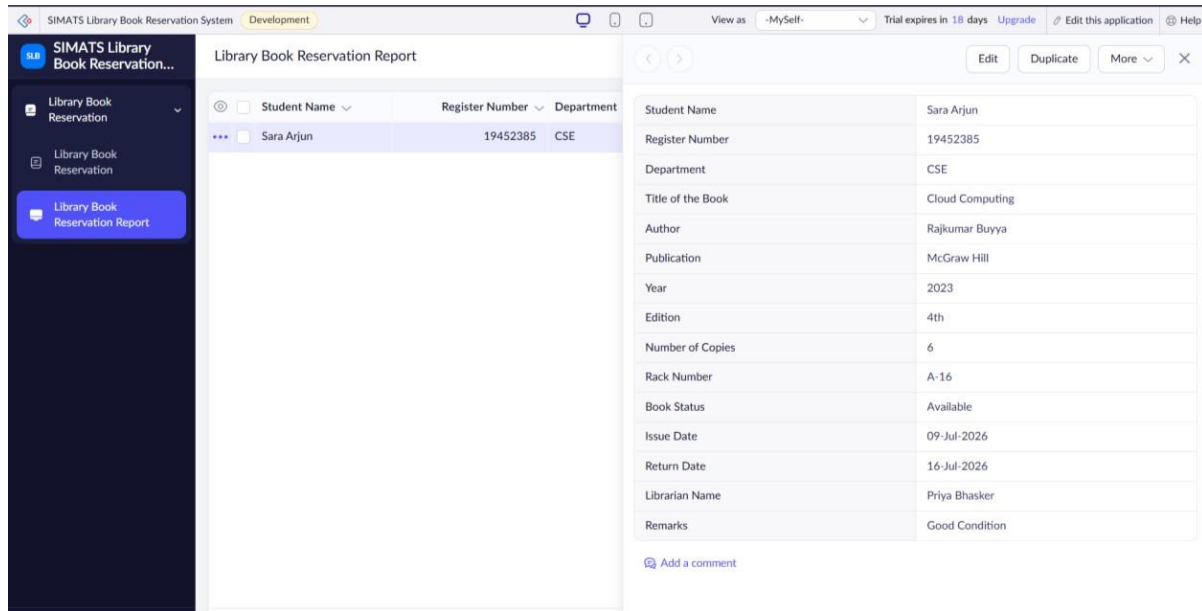
Library Book Reservation

Student Name *	
	First Name Last Name
Register Number *	
Department *	
Title of the Book *	
Author *	
	First Name Last Name
Publication	
Year *	
Edition	
Number of Copies *	
Rack Number	
Book Status *	
Issue Date *	

Library Book Reservation

Publication	
Year *	
Edition	
Number of Copies *	
Rack Number	
Book Status *	
Issue Date *	
Return Date *	
Librarian Name *	
	First Name Last Name
Remarks	

Submit



EXPERIMENT - 5: Mark Sheet Processing System

Theory: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student, Subjects, Marks, Average, percentage, rank, overall performance etc.

Aim: To create a simple cloud software application for a Mark Sheet Processing System using a Cloud Service Provider to demonstrate SaaS.

Algorithm:

1. Create a new application named "Mark Sheet Processing System" on the Cloud Service Provider.
2. Design a form with fields such as Student Name, Register Number, Department, Semester, Subject 1 to Subject 5 and Overall Performance.
3. Publish the form and enter sample subject marks for a student to test automatic calculation of Total Marks and Average.
4. Verify the calculated Total Marks, Average and Overall Performance in the Student Mark Sheet Report.

Result: A cloud-based Mark Sheet Processing System was successfully created, and sample student marks were entered with the Total, Average and Percentage computed and displayed correctly, demonstrating SaaS.

Screenshots:

The screenshot displays the 'Mark Sheet Processing System' interface. On the left is a dark sidebar with navigation options: 'Student Mark Sheet' (selected), 'Student Mark Sheet', and 'Student Mark Sheet Report'. The main area shows the 'Student Mark Sheet Report' for 'Sara Khan' with columns for 'Student Name', 'Register Number', and 'Department'. A table on the right lists subjects and marks:

Subject	Marks
Subject 1	95
Subject 2	88
Subject 3	90
Subject 4	85
Subject 5	92
Total Marks	450
Average	90.00
Overall Performance	

Below the table is a link to 'Add a comment'.

Theory: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org, over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Algorithm:

1. Create a new application named "Payroll Processing" on the Cloud Service Provider.

2. Design a form with fields such as Employee ID, Employee Name, Basic Pay, DA, HRA, CCA, PF, Income Tax, LOP, Gross Salary and Net Salary.
3. Publish the form and enter sample employee salary details to test the pay roll workflow.
4. Verify the computed Gross Salary and Net Salary in the Pay Roll Form Report.

Result: A cloud-based Payroll Processing System was successfully created, and sample employee payroll data was entered with Gross Pay and Net Pay calculated and verified, demonstrating SaaS.

Screenshots:

The top screenshot displays the 'Pay Roll form' interface. It features a sidebar with navigation options: 'Pay Roll form', 'Pay Roll form', and 'Pay Roll form Report'. The main form contains the following fields:

- Employee ID * (text input)
- Employee Name * (split into First Name and Last Name text inputs)
- Basic pay (text input)
- DA (text input)
- HRA (text input)
- CCA (text input)
- PF (text input)
- Income Tax (text input)
- LOP (text input)
- Gross Salary (text input)
- Net Salary (text input)

The bottom screenshot displays the 'Pay Roll form Report' interface. It shows a table with the following data:

Employee ID	Employee Name
124	tara kumar
123	sara ali

Below the table, a detailed breakdown of salary components is shown for the selected employee (Employee ID 124, Employee Name tara kumar):

Employee ID	124
Employee Name	tara kumar
DA	10
HRA	15
CCA	2500
PF	5
Income Tax	8
LOP	0
Gross Salary	65000
Net Salary	58500
Basic pay	50000

At the bottom of the report, there is a link to 'Add a comment'.

EXPERIMENT - 7: Student Attendance System

Theory: Create a simple cloud software application for Students Attendance System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the students, course, present, Absent, course attendance, OD request, grade, attendance percentage, etc.

Aim: To create a simple cloud software application for a Students Attendance System using a Cloud Service Provider to demonstrate SaaS.

Algorithm:

1. Create a new application named "Student Attendance System" on the Cloud Service Provider.
2. Design a form with fields such as Student ID, Student Name, Course, Date, Attendance Status (Present/Absent), Course Attendance, Total Classes, Attendance Percentage, OD Request/Reason, Grade and Faculty Name.
3. Publish the form and enter sample attendance records for a student to test the workflow.
4. Verify the calculated Attendance Percentage and grade in the Student Attendance Report.

Result: A cloud-based Student Attendance System was successfully created, and sample attendance data was entered with the attendance percentage computed and verified, demonstrating SaaS.

Screenshots:

The screenshot displays the 'Student Attendance System' web application. The interface includes a top navigation bar with the application name, a 'Development' status indicator, and user controls like 'View as -MySelf-' and 'Trial expires in 18 days'. A left sidebar menu contains options for 'Student Attendance' and 'Student Attendance Report'. The main content area, titled 'Student Attendance', contains a form with the following fields: 'Student ID' (text input with '#####'), 'Student Name' (split into 'First Name' and 'Last Name' text inputs), 'Course' (dropdown menu with '-Select-'), 'Date' (text input with 'dd-MMM-yyyy'), 'Attendance Status' (radio buttons for 'Present' and 'Absent'), 'Course Attendance' (text input with '#####'), 'Total Classes' (text input with '#####'), 'Attendance Percentage' (text input with '#####.##'), 'OD Request' (radio buttons for 'yes' and 'no'), and 'OD Reason' (text input). The form is designed with a clean, modern aesthetic using a light blue and white color scheme.

The screenshot displays a web application titled "Student Attendance System" in a development environment. The sidebar on the left contains the application logo and two menu items: "Student Attendance" (highlighted) and "Student Attendance Report". The main content area, titled "Student Attendance", features a form with the following fields:

- Course Attendance ***: A text input field containing "#####".
- Total Classes ***: A text input field containing "#####".
- Attendance Percentage**: A text input field containing "#####.###".
- OD Request ***: Radio buttons for "yes" and "no".
- OD Reason**: A text input field.
- Grade ***: A dropdown menu showing "-Select-".
- Faculty Name ***: Two text input fields for "First Name" and "Last Name".
- Remarks**: A large text input field.

EXPERIMENT - 8: Student Counselling System

Theory: Create a simple cloud web application for student counselling system using any Cloud Service Provider to demonstrate SaaS with the fields of student name, age, address, phone, subjects studied, Year of passing, percentage of marks, domain interest, etc.

Aim: To create a simple cloud web application for a Student Counselling System using a Cloud Service Provider to demonstrate SaaS.

Algorithm:

1. Create a new application named "Student Counselling System" on the Cloud Service Provider.
2. Design a form with fields such as Student Name, Age, Address, Phone Number, Email, Course Completed, Year of Passing, Percentage of Marks, Domain Interest, Counselling Date, Counsellor Name and Remarks.
3. Publish the form and enter sample student counselling details to test the workflow.
4. Verify the submitted record in the Student Counselling Form Report to confirm the data was stored correctly.

Result: A cloud-based Student Counselling System was successfully created, and sample student counselling data was entered and verified in the report, demonstrating SaaS.

Screenshots:

Theory: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Aim: To create a simple cloud software application for a Hospital Information System for SIMATS Hospital using a Cloud Service Provider to demonstrate SaaS.

Algorithm:

1. Create a new application named "SIMATS Hospital Information System" on the Cloud Service Provider.
2. Design a Patient Registration form with fields such as Patient Name, Age, Gender, Address, Phone, Email, Attender Name, Attender Phone, Doctor Name, Disease, Blood Group, Date of Admission, Ward Type and Consultation Fee.
3. Publish the form and enter sample patient registration details to test the workflow.
4. Verify the submitted patient record in the Patient Registration Report.

Result: A cloud-based Hospital Information System for SIMATS Hospital was successfully created, and a sample patient registration record was stored and verified, demonstrating SaaS.

Screenshots:

The screenshot displays the 'SIMATS Hospital Information System' web application. The interface includes a dark sidebar on the left with a logo and navigation links: 'Patient Registration' (selected), 'Patient Registration Report', and 'Patient Registration'. The main content area is titled 'Patient Registration' and contains a form with the following fields: 'Patient Name *' (with a 'First Name' label), 'Age *' (with a '#####' placeholder), 'Gender *' (a dropdown menu showing '-Select-'), 'Address *' (a large text area), 'Phone *' (with a country code dropdown set to '+91' and a number field containing '81234 56789'), 'Email *' (a text field), 'Attender Name *' (with a 'First Name' label), 'Attender Phone *' (with a country code dropdown set to '+91' and a number field containing '81234 56789'), 'Doctor Name *' (a dropdown menu showing '-Select-'), and 'Disease *' (a text field). The top of the application shows a header with the title 'SIMATS Hospital Information System', a trial expiration notice 'Trial expires in 17 days', and links for 'Upgrade', 'Edit this application', and 'Help'.

SIMATS Hospital Information System
Trial expires in 17 days
Upgrade
Edit this application
Help

SIMATS Hospital Information System
Patient Registration
Patient Registration Report

Patient Registration

Attender Name *

First Name

Attender Phone *

+91 81234 56789

Doctor Name *

-Select-

Disease *

Blood Group *

-Select-

Date of Admission *

dd-MMM-yyyy

Ward Type *

-Select-

Consultation Fee *

###,###,##

Remarks

Submit

Patient Registration Report

Patient Name	Age	Gender	Address
Oviya	20	Female	ABC Flats, Vadapalani, Chennai-600093

Edit
Duplicate
More
X

Patient Name	Oviya
Age	20
Gender	Female
Address	ABC Flats, Vadapalani, Chennai-600093
Phone	+91789456120
Email	oviya@gmail.com
Attender Name	Sara
Attender Phone	+917984561230
Doctor Name	Dr. Meera
Disease	Viral Fever
Blood Group	A+
Date of Admission	22-Jul-2026
Ward Type	General Ward
Consultation Fee	₹ 5,000.00
Remarks	good

Add a comment

EXPERIMENT - 10: Student Information System

Theory: Create a simple cloud software application to run a c program to display the student information using any Cloud Service Provider to demonstrate SaaS with the template of student name, reg no, address, phone, age, courses, grades and attendance report, progress report, Semester mark sheet forms.

Aim: To create a simple cloud software application to display student information using a Cloud Service Provider to demonstrate SaaS, including Progress Report and Semester Mark Sheet forms.

Algorithm:

1. Create a new application named "Student Information System" on the Cloud Service Provider with three linked forms: Student Information, Progress Report and Semester Mark Sheet.

2. Design the Student Information form with fields such as Student Name, Register Number, Address, Phone Number, Age, Department, Semester, Courses, Grades and Attendance Percentage.
3. Design the Progress Report form (Internal Marks, Assignment Marks, Lab Marks, Overall Progress, Faculty Remarks) and the Semester Mark Sheet form (Subject 1-5, Total Marks, Percentage, Grade), linked by Register Number.
4. Publish the application, enter sample data in all three forms, and verify the records in their respective reports.

Result: A cloud-based Student Information System with linked Progress Report and Semester Mark Sheet forms was successfully created, and sample student records were stored and verified across all three reports, demonstrating SaaS.

Screenshots:

The top screenshot displays the 'Student Information' form. It includes the following fields:

- Student Name * (First Name, Last Name)
- Register Number * (#####)
- Address *
- Phone Number * (+91 81234 56789)
- Age * (#####)
- Department * (-Select-)
- Semester * (-Select-)
- Courses *
- Grades *
- Attendance Percentage * (#####)

The bottom screenshot displays the 'Student Information Report' table. It includes the following columns:

- Student Name
- Register Number
- Address
- Phone Number
- Age
- Department
- Semester
- Courses
- Grades
- Attendance Percentage

Student Name	Register Number	Address	Phone Number	Age	Department	Semester	Courses	Grades	Attendance Percentage
Oviya Prasannakumar	192324000	ABC Flats, Vadapalani, Chennai-93	+917894561230	20	AI&DS	Semester 5	DBMS	A	98.00

Student Information System

Progress Report

Student Information

Progress Report

Progress Report Report

Progress Report Report

Semester Mark Sheet

Register Number *

Internal Marks *

Assignment Marks *

Lab Marks *

Overall Progress *

Faculty Remarks *

Submit

Student Information System

Progress Report Report

Register Number

Internal Marks

192324000

92

Register Number

Internal Marks

Assignment Marks

Lab Marks

Overall Progress

Faculty Remarks

192324000

92

90

89

Good

good

Add a comment

Student Information System

Semester Mark Sheet

Register Number *

Subject 1 *

Subject 2 *

Subject 3 *

Subject 4 *

Subject 5 *

Total Marks

Percentage

Grade *

[Submit](#)

Student Information System

Semester Mark Sheet Report

Register Number	Subject 1
192324000	85

Register Number	192324000
Subject 1	85
Subject 2	90
Subject 3	80
Subject 4	88
Subject 5	92
Total Marks	435
Percentage	87.00
Grade	A

[Add a comment](#)

EXPERIMENT - 11: Online Super Market

Theory: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Aim: To create a simple cloud software application for an Online Super Market using a Cloud Service Provider to demonstrate SaaS, including an Item Classification form.

Algorithm:

1. Create a new application named "Online Super Market" on the Cloud Service Provider with two linked forms: Customer Order and Item Classification.

2. Design the Customer Order form with fields such as Customer Name, Address, Phone Number, Email, Item Category, Item Name, Quantity, Price, Total Amount and Payment Method.
3. Design the Item Classification form with fields such as Item Name, Category, Brand, Price, Stock Quantity, Supplier Name, Created At/By and Last Updated At/By.
4. Publish the application, enter sample customer order and item classification records, and verify them in the respective report sections.

Result: A cloud-based Online Super Market application with Customer Order and Item Classification forms was successfully created, and sample order and item data were stored and verified, demonstrating SaaS.

Screenshots:

Online Super Market

Customer Order

Customer Name *

Address *

Phone Number * +91 81234 56789

Email *

Item Category * -Select-

Item Name *

Quantity * #####

Price * #####.##

Total Amount #####.##

Payment Method * -Select-

Submit

Online Super Market

Customer Order Report

Customer Name	Address	Phone Number
Oviya	ABC flats, Vadapalani, Chennai-99	+917894561230

Customer Name: Oviya

Address: ABC flats, Vadapalani, Chennai-99

Phone Number: +917894561230

Email: oviya@gmail.com

Item Category: Dairy Products

Item Name: curd

Quantity: 1

Price: \$ 30.00

Payment Method: Cash

Total Amount: 30.00

Add a comment

Online Super Market

Customer Order

Item Classification

Item Classification

Item Classification Report

Item Classification

Item Name *

Category *

Brand *

Price *

Stock Quantity *

Supplier Name *

Created At *

Updated At *

Created By *

Last Updated By *

Submit

Online Super Market

Customer Order

Item Classification

Item Classification

Item Classification Report

Item Classification Report

Item Name

Category

Brand

Milk Biscuits

Snacks

Britannia

Item Name

Category

Brand

Price

Stock Quantity

Supplier Name

Created At

Updated At

Created By

Last Updated By

Milk Biscuits

Snacks

Britannia

\$ 20.00

100

Britannia

01-Jul-2026 21:03:39

08-Aug-2026 21:03:43

abcportal66

abcportal66

Add a comment